

When More Is Less: Sensory Interaction Load in Multisensory Virtual Reality

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Abstract

Virtual Reality (VR) experiences can be enriched through multi-sensory stimulation, with the risk of increasing processing demands. Drawing on the environmental psychology framework and grounded in sensory overload theory, this exploratory 2×2 between-subjects VR experiment with 132 participants investigates how olfactory cues and interaction dynamics jointly influence psychological, physiological, and behavioral responses. Results reveal interaction effects on pleasure and usability, with dynamic scented scenarios associated with lower ratings. Conscious scent perception increased product choice, while dynamic interaction reduced scent awareness. Overall, the findings suggest the need of balanced multisensory VR design.

CCS Concepts

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

Keywords

virtual reality, multi-sensory, sensory–interaction load, psychological, physiological, behavior, laboratory-study

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1 INTRODUCTION

Virtual Reality (VR) enables highly immersive experiences, which can be enhanced through multisensory stimulation [6]. Multisensory stimulation refers to the deliberate activation of multiple sensory modalities in order to intensify and enrich the experience [7]. Drawing on the environmental psychology framework by Mehrabian and Russell, environmental stimuli - such as scented or interactive virtual environments - are assumed to influence users'

emotional experience (Pleasure-Arousal-Dominance; PAD), which in turn shape behavioral responses [5].

However, combining multiple sensory cues does not necessarily lead to more positive experiences. Prior research suggests that additional sensory input may increase processing demands and potentially impair the experience [4, 9]. Rather than conceptualizing these effects as overstimulation in a strict clinical sense, which describes a condition in which sensory demands exceed users' processing capacity, potentially resulting in sensory overload and diminished experiential outcomes, they may be better understood in terms of sensory–interaction load, defined as the combined attentional and cognitive demands imposed by multisensory input and active interaction.

From an HCI perspective, this raises a critical design challenge: as VR systems increasingly combine rich interaction techniques with multisensory feedback, designers lack empirical guidance on how interaction complexity and sensory augmentation jointly shape user experience. While prior work often studies sensory modalities in isolation, less is known about their combined cognitive and attentional demands during active interaction.

Against this background, the present exploratory study investigates how olfactory stimulation and interaction dynamics in VR jointly affect emotional experience (psychological, physiological) and behavioral responses [2]. By systematically manipulating scent presence and interaction mode in a controlled VR environment, this research contributes empirical evidence to ongoing discussions on multisensory VR design and its limits. In this context, interaction dynamics refer to the degree of active user manipulation and movement within the virtual environment (i.e., grasping and inspecting objects), rather than to temporal variations in scent diffusion.

2 METHODOLOGY

A 2×2 between-subjects laboratory design (scent: absent vs. present × interaction: static vs. dynamic) was implemented in Virtual Reality in 2025 with 132 participants. In addition to psychological self-report measures, physiological data were collected to complement and validate experiential responses. The self-developed virtual environment featured four fruits (apple, pear, banana, orange) placed on tables (see fig.1). After completing a brief test dynamic scenario to ensure technical understanding, participants experienced the study-relevant environment for three minutes and were instructed to choose one fruit they could take home in real life.



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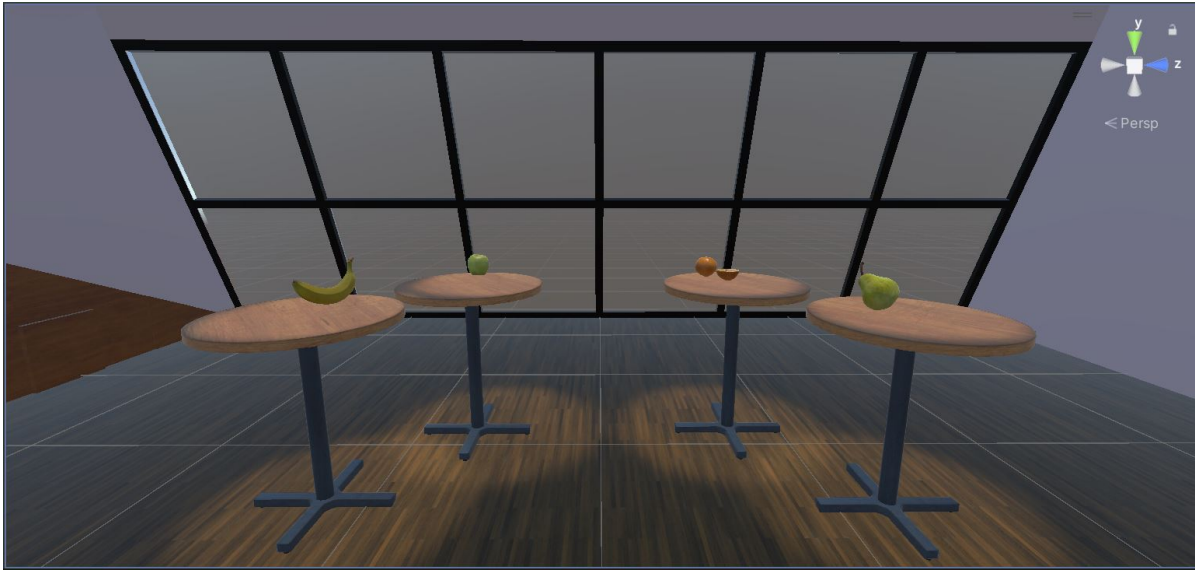


Figure 1: Self-created VR-scenario for laboratory study



Figure 2: Study equipment: VR-device, Galvic Skin Response and ambient diffusion system (left to right)

Interaction was manipulated such that participants either observe the fruits from a fixed position (static condition) or actively move, grasp and inspect them (dynamic condition), using one controller, while the other hand was equipped with the GSR sensor for physiological recording. Each interaction condition was presented either without scent or with a scenario congruent orange scent, diffused continuously from the start of the VR experience using a room-based ambient diffusion system positioned behind the participant. Scent was implemented as an ambient environmental cue rather than as an object-specific stimulus, as the study aimed to influence fruit choice rather than simulate localized odor sources. The diffusion rate was calibrated in pretests to ensure perceptibility without inducing irritation or saturation. To ensure a stable ambient scent level, the diffusion system automatically interrupted scent emission at regular intervals, preventing continuous buildup and supporting a constant room-level olfactory stimulation throughout the exposure. The four experimental conditions were: (1) static

without scent ($n = 35$), (2) static with scent ($n = 37$), (3) dynamic without scent ($n = 29$) and (4) dynamic with scent ($n = 31$).

The measured variables are included in the framework (see figure 3). Prior to the VR exposure, participants reported sociodemographic information, potential fruit and scent allergies and current mood (MSF; $\alpha = .70$, with one item deleted) [8]. Emotional experience was operationalized by (1) Pleasure–Arousal–Dominance (PAD; KMO = .8; Bartlett’s Test: $\chi^2(153) = 857.1$, $p \leq .001$) [5], (2) User Experience (UE; KMO = .7; Bartlett’s Test: $\chi^2(28) = 394.5$, $p \leq .001$) [11], (3) Sense of Presence (P; KMO = .8; Bartlett’s Test: $\chi^2(91) = 684.8$, $p \leq .001$) [12], (4) Cybersickness (CS, KMO = .7; Bartlett’s Test: $\chi^2(15) = 163.8$, $p \leq .001$) [3]. (5) Galvanic Skin Response (GSR- μS) and (6) Heart Rate (HR) were recorded using a Shimmer 3 GSR+ system with palmar electrodes and a Polar H10 chest strap sensor. Physiological measures are reported to support interpretation of experiential patterns rather than to infer discrete emotional states. Behavioral response was operationalized by product-related attitudes toward the chosen fruit and

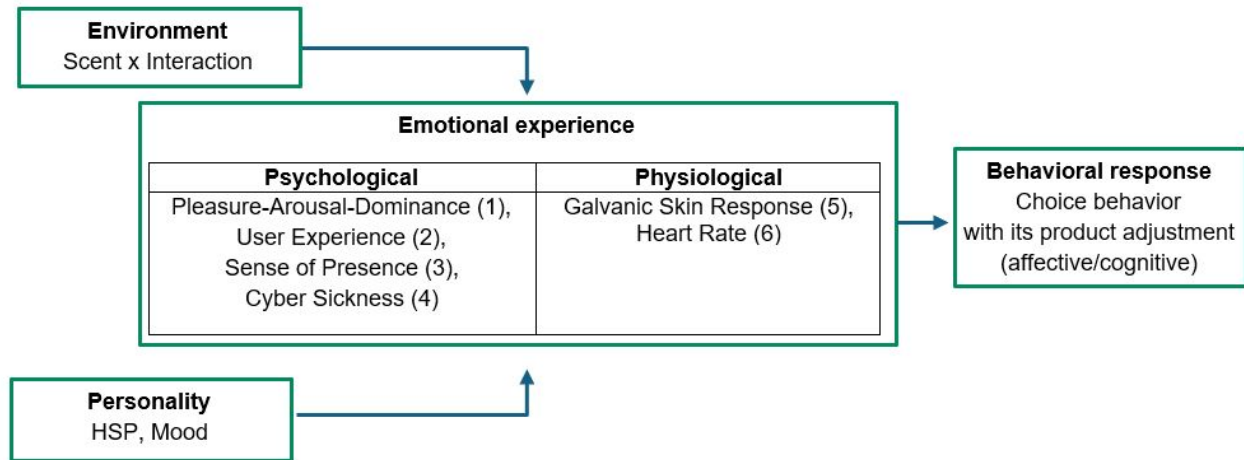


Figure 3: Environmental psychological framework; recreated from [5] and measured variables of the VR study

toward the orange by unidimensional items capturing affective and cognitive components. Behavioral response was operationalized as fruit choice. Additionally, sensory processing sensitivity was assessed using the High Sensory Processing scale (HSP; $\alpha = .79$) [10] and scent perception (pleasantness, familiarity, intensity, and perceived congruence) was measured on 0–10 scales [1]. Given the exploratory nature of the study, statistical analyses focus on identifying patterns and interaction effects rather than on hypothesis confirmation.

The final sample included 132 participants, of whom complete data were available for 126 individuals for heart rate and 131 for GSR measurements. None had scent-related allergies. The sample was 55% female, had a mean age of 27 years ($SD = 10.3$) and was predominantly highly educated, with 87% holding a university degree. Participants reported relatively low prior VR experience ($M = 3.1$, $SD = 2.8$). The majority reported no current nasal congestion (93%) and no smoking behavior (81%). These variables were not excluded from analysis, as preliminary tests indicated no systematic effects. Overall, 47 participants reported having perceived a scent; of these, 44 were able to evaluate it (static scenario: 28; dynamic scenario: 16). The scent was rated as pleasant ($M = 7.4$, $SD = 2.4$), familiar ($M = 7.1$, $SD = 2.2$), of moderate intensity ($M = 7.0$, $SD = 2.5$) and appropriate for the situation ($M = 7.3$, $SD = 2.3$).

3 FINDINGS

3.1 Psychological and psychological responses for emotional experience

A between-subjects ANOVA revealed a significant main effect of interaction on arousal ($F(1,128) = 5.51$, $p = .020$) and realism ($F(1,128) = 3.94$, $p = .049$), indicating that dynamic VR environments elicited higher arousal levels and greater perceived realism than static environments. The dynamic condition also yielded significant effects in physiological measures. Participants in the dynamic scenario exhibited a higher average heart rate than those in the static scenario, potentially reflecting increased physical movement. This difference

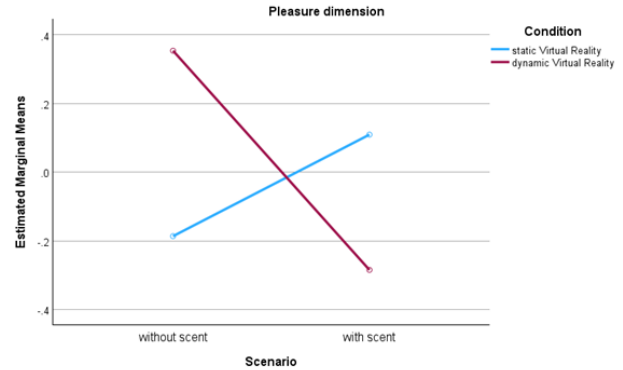


Figure 4: Estimated Marginal Means of Pleasure dimension

was significant (dynamic: $M = 89$; static: $M = 83$; $F(1,125) = 4.70$, $p = .032$) and was accompanied by a significant difference in average inter-beat interval (IBI; $F(1,125) = 4.66$, $p = .033$). Moreover, a significant interaction effect between scent and interaction condition was observed for pleasure, with participants in the scented dynamic condition perceiving the experience as less pleasurable ($F(1,128) = 5.73$, $p = .018$) (see fig.4). Even though a significant interaction for one of the two user experience factors - perceived usability and understandability, $F(1,128) = 5.28$, $p = .023$, where participants in the scented dynamic scenario reported significantly lower usability and understandability, perceive less realism $F(1,128) = 4.52$, $p = .035$ and had higher average peak amplitude of galvanic skin response $F(1,127) = 4.96$, $p = .028$. The analyses revealed no significant main or interaction effects for Presence or Cyber Sickness.

3.2 Behavioral response

Regarding product attitudes, measured unidimensionally on affective and cognitive dimensions, the interaction term revealed a significant main effect of scent on the cognitive evaluation of the

orange, $F(1,128) = 6.11$, $p = .015$. The chi-square test revealed no significant association between the categorical variables of scented condition and the orange choice (Pearson's $\chi^2(1) = 2.77$, $p = .096$), indicating only a non-significant trend. Crucially, participants who consciously perceived the scent were significantly more likely to choose the orange, Pearson's $\chi^2(1) = 9.42$, $p = .002$, with Odds Ratio approximately three times higher ($OR \approx 3.27$). This finding underscores that multisensory cues in VR do not operate implicitly by default but require sufficient attentional resources to reach conscious awareness and influence behavior. In the scented condition a conscious scent detection was less frequent in the scented dynamic scenario, $\chi^2 = 4.28$, $p = .039$.

3.3 Influence of personality characteristics

The effects of High Sensory Processing (HSP) and mood on the dependent psychological variables (PAD, P, CS, UE) were analyzed. While HSP sensitivity showed no general main effect or moderation effects, mood significantly influenced several dependent variables as a main effect (e.g., Pleasure, Spatial Presence, CS, both dimensions of UE) and moderated the effects of the experimental conditions only for CS - visual strain & balance disturbances, with no significant moderation observed for the other measures.

4 CONCLUSION

Overall, the results indicate that adding multiple sensory modalities does not automatically enhance immersive experiences. From a cognitive load perspective, dynamic interaction likely increased intrinsic and extraneous processing demands, leaving fewer attentional resources available for olfactory perception. This aligns with cognitive load theory, which posits that sensory information competing for limited working memory capacity may fail to reach conscious awareness. The findings therefore suggest that olfactory cues in VR are particularly susceptible to attentional suppression under high interaction load.

The decrease in pleasure observed in the scented dynamic condition, combined with increased arousal and reduced usability, may reflect heightened sensory–interaction load. Actively navigating and manipulating objects in VR likely increases attentional demands, thereby reducing available cognitive resources for processing olfactory cues. As a result, scent perception may be masked when interaction complexity is high.

The behavioral findings further underline the importance of conscious scent perception. Importantly, when scent reached conscious awareness, it significantly increased product choice likelihood. This indicates that olfactory cues can be effective behavioral nudges in VR—provided sufficient attentional capacity and perceptual salience are ensured.

The effects were consistent across levels of sensory sensitivity but moderated by mood, highlighting mood as a key factor in multisensory VR experiences.

While the present study focuses on olfactory cues, the concept of sensory–interaction load may extend to other sensory modalities such as haptic, auditory, or thermal feedback. Any additional modality that competes for attentional resources during complex interaction may similarly risk diminishing experiential quality. Future

research should therefore systematically compare modality-specific load effects to better inform multisensory VR design strategies.

Based on the findings, design-oriented insights for multisensory VR systems derive: (1) More sensory input is not always better: Olfactory cues can impair experience when combined with high interaction complexity. (2) Interaction moderates sensory effectiveness: Designers should evaluate sensory augmentation in relation to interaction complexity, particularly in tasks involving object manipulation and navigation. (3) Align scent with interaction semantics: Ambient scents should be carefully matched to interactive objects to avoid incongruence. (4) Support scent awareness: If scent is intended to influence behavior, designs should ensure sufficient attentional capacity for olfactory processing.

A key limitation concerns the ambient implementation of scent. The continuous diffusion of an orange odor, regardless of the currently inspected object, may have introduced crossmodal incongruence, potentially affecting pleasure and usability ratings independently of attentional load. Physiological measures should be interpreted cautiously due to between-subject variability and potential movement effects. Future work should incorporate within-subject baselines, movement-controlled conditions, and object-specific olfactory cues. Longer exposure times could further clarify the role of olfactory adaptation in interactive VR. Future studies should compare ambient versus object-specific scent diffusion to disentangle sensory–interaction load from crossmodal congruence effects.

This exploratory study demonstrates that olfactory stimulation does not universally enhance VR experiences. Instead, its effects depend critically on interaction dynamics and attentional demands. The results suggest that olfactory cues in VR are particularly sensitive to attentional competition and may be among the first modalities to be suppressed under high interaction load. By highlighting sensory–interaction load as a key factor, this work contributes to a more nuanced understanding of multisensory VR design and offers practical guidance for interaction design, immersive systems, and experiential evaluation, emphasizing that additional sensory input should be carefully and strategically integrated to optimize user experience.

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